

Comprehensive Summary: Abstract Algebra, Galois Theory, and Constructibility

1 Field Extensions

Question 1. What is the degree of the extension $\mathbb{R}/\mathbb{Q}$?

The degree $[\mathbb{R} : \mathbb{Q}]$ is infinite. This is because $\mathbb{R}$ contains transcendental numbers (like π and e), which are not roots of any polynomial with rational coefficients. Since it contains elements not algebraic over $\mathbb{Q}$, the extension cannot be finite.

Question 2. Does $[E : K]$ being finite imply E/K is algebraic?

Yes. If $[E : K] = n$, then for any $\alpha \in E$, the $n + 1$ elements $\{1, \alpha, \alpha^2, \dots, \alpha^n\}$ must be linearly dependent over K . This linear dependence yields a non-zero polynomial $p(x) \in K[x]$ having α as a root.

Question 3. Does an algebraic extension imply a finite extension?

No. A standard counterexample is the algebraic closure of the rationals, $\bar{\mathbb{Q}}$, over $\mathbb{Q}$. It contains sub-extensions $\mathbb{Q}(\sqrt[d]{2})$ of arbitrary degree d , meaning the total degree is infinite, despite every single element being algebraic over $\mathbb{Q}$.

Question 4. If all elements of an extension have a minimal polynomial of at most degree n , is the extension finite?

In general, this is false. A counterexample is the infinite inseparable extension $\mathbb{F}_p(t_1, t_2, \dots)/\mathbb{F}_p(t_1^p, t_2^p, \dots)$, where every element has a minimal polynomial of degree at most p . However, if the extension is strictly *separable*, the statement becomes true (proven via the Primitive Element Theorem).

2 Galois Theory and Automorphisms

Question 5. Does $[E : K] = 2$ imply E/K is normal?

Yes. For any $\alpha \in E \setminus K$, the sub-extension degree $[K(\alpha) : K]$ must divide 2, so $E = K(\alpha)$. The minimal polynomial of α is a quadratic $x^2 + bx + c$. The second root is strictly $-b - \alpha$, which is also in E . Thus, E is the full splitting field of this polynomial, making it a normal extension.

Question 6. Why does a polynomial in the base field remain invariant over an automorphism fixing the base field?

An automorphism σ fixing K maps every coefficient $c_i \in K$ to itself ($\sigma(c_i) = c_i$). Because σ is a field homomorphism, applying it to the equation $f(\alpha) = 0$ yields $f(\sigma(\alpha)) = \sigma(0) = 0$. This proves automorphisms map roots to other roots of the same polynomial.

Question 7. If E/K is Galois and F is an intermediate field, is E/F Galois?

Yes. E/K being Galois means it is separable and normal. E/F inherits separability because the minimal polynomial over F divides the minimal polynomial over K . It inherits normality because E remains the splitting field of the same polynomial, now viewed over F .

Question 8. For finite fields $K \subseteq F \subseteq E$, is F/K Galois given E/K is Galois?

Yes. For finite fields, the Galois group $G = \text{Gal}(E/K)$ is cyclic, and therefore abelian. Subgroups of abelian groups are always normal. By the Fundamental Theorem of Galois Theory, since the subgroup corresponding to F is normal, F/K is a Galois extension. (Alternatively, all finite extensions of finite fields are separable and normal inherently).

3 Cyclotomic Extensions

Question 9. Is $\mathbb{Q}(e^{2\pi i/n})/\mathbb{Q}$ a separable extension?

Yes. $\mathbb{Q}$ has characteristic 0, meaning it is a perfect field, so all its algebraic extensions are separable. Alternatively, the root ζ_n satisfies $x^n - 1 = 0$, and the derivative nx^{n-1} shares no common roots with $x^n - 1$, ensuring distinct roots.

Question 10. What is the degree of the extension $\mathbb{Q}(e^{2\pi i/n})/\mathbb{Q}$?

The degree is exactly Euler's totient function, $\phi(n)$. It corresponds to the degree of the irreducible n -th cyclotomic polynomial $\Phi_n(x)$.

Question 11. What is the cyclotomic polynomial for a prime p ?

For a prime p , $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$.

Question 12. Is $\mathbb{Q}(e^{2\pi i/p})/\mathbb{Q}$ a normal extension?

Yes. It is the splitting field of $x^p - 1$ over $\mathbb{Q}$. The roots are $1, \zeta_p, \zeta_p^2, \dots, \zeta_p^{p-1}$, all of which are generated by ζ_p and thus strictly lie in the field $\mathbb{Q}(\zeta_p)$.

4 Group Theory

Question 13. Let G be a finite abelian group of order n . If $|\{x \in G : x^d = 1\}| \leq d$ for all $d|n$, is G cyclic?

Yes. The proof partitions the group elements by exact order d (count is $\psi(d)$). The condition forces $\psi(d) \leq \phi(d)$. Since $\sum_{d|n} \psi(d) = n$ and $\sum_{d|n} \phi(d) = n$, we must have $\psi(d) = \phi(d)$ for all divisors. Thus, $\psi(n) = \phi(n) \geq 1$, proving an element of order n exists.

Example: This applies to the multiplicative group of any finite field F^* , proving it is cyclic.

Counterexample: It fails for the Klein four-group $(\mathbb{Z}_2 \times \mathbb{Z}_2)$, where $d = 2$ yields 4 solutions to $2x = 0$, violating the condition $4 \leq 2$.

Question 14. In a cyclic group, is the number of elements of order d less than or equal to d ?

Yes. In a cyclic group, there is exactly one subgroup of order d . The elements of order d are precisely the generators of this subgroup, of which there are $\phi(d)$. Because $\phi(d) \leq d$ holds universally, the statement is true.

Question 15. Can every finite abelian group be isomorphic to a quotient group of $(\mathbb{Z}/n\mathbb{Z})^\times$?

Yes. Decompose the group into $\mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_k}$. By Dirichlet's Theorem, pick distinct primes $p_i \equiv 1 \pmod{d_i}$. Then d_i divides $p_i - 1$, allowing a surjective homomorphism from $\mathbb{Z}_{p_i-1}$ to $\mathbb{Z}_{d_i}$. Taking $n = p_1 \cdots p_k$, the Chinese Remainder Theorem provides the required surjective homomorphism from $(\mathbb{Z}/n\mathbb{Z})^\times$, and the First Isomorphism Theorem yields the final quotient structure.

5 Constructible Numbers

Question 16. What is the geometric and algebraic definition of constructible numbers?

Geometrically, the set of constructible numbers Q_∞ is formed step-by-step from $\{0, 1\}$ via intersections of Lines (LL), Lines and Circles (LC), and Circles (CC): $Q_n = Q_{n-1} \cup Q_{n-1}(LL) \cup Q_{n-1}(LC) \cup Q_{n-1}(CC)$. Algebraically, a number is constructible iff it belongs to a field at the top of a tower of quadratic extensions over $\mathbb{Q}$. This mandates that its minimal polynomial over $\mathbb{Q}$ must have a degree that is a power of 2 (2^k). This algebraic bridge proves classical problems (doubling the cube, trisecting arbitrary angles, squaring the circle) are impossible.

Question 17. Is a complex number $z = x + iy$ constructible iff $\text{Re}(z)$ and $\text{Im}(z)$ are both constructible?

Yes. The constructible numbers $\mathbb{K}$ form a field containing i . If $x, y \in \mathbb{K}$, then $x + iy \in \mathbb{K}$ due to closure under addition and multiplication. Conversely, if $z \in \mathbb{K}$, since $\mathbb{K}$ is closed under complex conjugation, $\bar{z} \in \mathbb{K}$, allowing algebraic recovery of $x = \frac{z + \bar{z}}{2}$ and $y = \frac{z - \bar{z}}{2i}$.

Question 18. Is the regular 2^r -gon constructible?

Yes, for $r \geq 2$. Geometrically, this is achieved by repeatedly bisecting angles starting from a square. Algebraically, constructing a 2^r -gon equates to finding the primitive 2^r -th root of unity ζ_{2^r} . The degree is $[\mathbb{Q}(\zeta_{2^r}) : \mathbb{Q}] = \phi(2^r) = 2^{r-1}$, which is exactly a power of 2, thus satisfying the field-theoretic condition for constructibility.